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EXPERIMENT 8

Experiment 8: Conduct Z test for the Data Given using Python

AIM:

To test whether the average weight of a species of birds differs from 160 grams.

PROCEDURE:

1. Null Hypothesis (H_0): The average weight of the birds is 160 grams.
2. Alternative Hypothesis (H_1): The average weight of the birds is not 160 grams.
3. Sample: Measure the weights of 30 birds randomly selected from the population.
4. Z-Test: Conduct a Z-test to compare the sample mean to 160 grams.
5. Decision Rule: Use a significance level of $\alpha = 0.05$.

CODE:

```
import numpy as np
import scipy.stats as stats
# Define the sample data (hypothetical weights in grams)
sample_data = np.array([
    162, 158, 161, 159, 157, 163, 160, 158, 162, 159,
    161, 160, 159, 162, 161, 158, 160, 162, 159, 160,
    158, 163, 161, 160, 159, 162, 158, 161, 160, 163
])
# Population mean under the null hypothesis
population_mean = 160
# Calculate sample statistics
sample_mean = np.mean(sample_data)
sample_std = np.std(sample_data, ddof=1)
# Number of observations
n = len(sample_data)
# Calculate the Z-statistic
z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))
# Calculate the p-value
p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic)))
```

```
# Print results
print(f"Sample Mean: {sample_mean:.2f}")
print(f"Sample Standard Deviation: {sample_std:.2f}")
print(f"Z-Statistic: {z_statistic:.4f}")
print(f"P-Value: {p_value:.4f}")
# Decision based on the significance level
alpha = 0.05
if p_value < alpha:
    print("Reject the null hypothesis: The average weight is significantly different from 160 grams.")
else:
    print("Fail to reject the null hypothesis: There is no significant difference in average weight from 160 grams.")
```

OUTPUT:

```
Sample Mean: 160.20
Sample Standard Deviation: 1.71
Z-Statistic: 0.6406
P-Value: 0.5218
Fail to reject the null hypothesis: There is no significant difference in average weight from 160 grams.
```

RESULT:

The Z-test was conducted and the P-value was found to be **0.5218**. Since P-value > 0.05, the null hypothesis is accepted, and there is no significant difference from **160 grams**.